

Nicholas Hamilton

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EDUCATION

Bachelor of Arts in Computer Science

Berea College | 3.75 GPA

Expected Graduation: May 2026

SKILLS

Languages Proficient Python(3yrs) · SQL(3yrs) Intermediate JavaScript(2yrs) · TypeScript(2yrs) Beginner C++(2yrs) · Go(1yr)
Software Git · FastAPI · Flask · Docker · Ollama · Langchain · TensorFlow · Nest.js · React.js · Bash · Jupyter Notebook

EXPERIENCE

Software Engineering Intern

Jun 2025 - Aug 2025

[NextGen Justice](#)

San Jose, CA

- Orchestrated end-to-end modular RAG pipeline, from embedding to multiple vector store retrieval routes
- Authored the database schema using PostgreSQL, pgvector, and Prisma ORM, and implemented HNSW indexing
- Produced an AI chat bot, allowing for persistent memory with sliding window and summaries and source retrieval
- Secured the B2B environment by establishing authentication and authorization with role-based access
- Optimized cloud infrastructure integrating AWS S3 for document storage with on-premises LLM hosting in Docker

Lead Teacher's Assistant

Jan 2024 - May 2025

[Berea College](#)

Berea, KY

- Managed over 20 Teaching Assistants, creating lab schedules, leading weekly meetings, and facilitating with staff
- Automated bulk [student repo checkout/update](#) via Python and Bash, using git CLI calls to reduce review time by 30%
- Graded and tutored 15 hrs weekly for introductory programming, data structures & algorithms, and capstone projects
- Formulated individual career development plans for team members, fostering professional growth amongst TAs

Data Science Intern

May 2024 - July 2024

[Virginia Tech Data Science for the Public Good](#)

Blacksburg, VA

- Constructed data pipeline processing 50+ socioeconomic variables across continental United States
- Trained a model with XGBoost using R to project food insecurity levels throughout 3000+ counties
- Applied ARIMA time series forecasting to project socioeconomic variables through 2027 for resource allocation
- Programmed responsive UI for geospatial mapping with real-time filtering capabilities for 400+ food resource locations
- Established data quality controls addressing missing values, population thresholds, and geographic entity changes
- Presented findings demonstrating potential 15,252 reduction in food insecure individuals if our goals were achieved

Full Stack Engineer

Jan 2024 - May 2024

[UP Initiative](#)

Berea, KY

- Led 4-person team to build full-stack web application for local nonprofit serving 200+ unhoused individuals
- Architected database schema with complex relationships, achieving 4NF normalization, and displayed reactive queries
- Configured comprehensive visit logging system with complex SQL joins across multiple tables, generating detailed reports

PROJECTS

Data Analysis Journal | Personal Project (~50 hours) - [View Project](#)

Feb 2025

- Developed mobile-first Progressive Web Application with centralized state management and SPA routing
- Engineered a centralized analytics engine caching 7+ types of mood and productivity metrics for real-time user insights

CS Departmental Website | Team Project (~120 hours) - [GitHub Link](#)

Sep 2024 - Feb 2025

- Directed 15+ developer team across 3 committees establishing online information hub for 1,500+ students
- Implemented responsive React SPA with CSV-driven content management and real-time TA scheduling system
- Mentored team on RESTful API design, Git production workflows, and asynchronous Python domain driven architecture

Census Data GUI | Personal Project (~80 hours) - [GitHub Link](#)

Sep 2024 - Oct 2024

- Produced full-stack Census data platform with Flask integrating Census Bureau APIs for multi-table data processing
- Integrated authentication system with bcrypt hashing, rate limiting, and session management with audit logging
- Structured PostgreSQL database with specialized managers for user accounts, project organization, and security monitoring

Network Traffic Analysis | Team Project (~30 hours) - [GitHub Link](#) [View Project](#)

Sep 2024 - Oct 2024

- Created a network traffic analysis dashboard using Python, Dash, and Plotly, processing over 1 million packets
- Formulated Isolation Forest algorithm to detect network anomalies, visualizing results and presenting solutions
- Built data processing pipeline using Scapy and Pandas to analyze packet captures, reducing analysis time by over 50%

Activities

[Virginia Cooperative Extension](#) - Co-author of peer-reviewed publication covering food insecurity

May 2025

Student Government Association - Liaison for the Non-Traditional Student Association (NTSA)

Aug 2023 - May 2024

Campus Improvement Committee - Vice-Chairperson overseeing campus events and drafting budgets

Aug 2023 - May 2024